King Salman International University

Faculty of Computer Science and Engineering

Project Title

Intelligent Robot for Flame Location Detection

A Thesis Submitted to Computer Engineering Department
Faculty of Computer Science & Engineering, KSIU

In Partial Fulfillment of **B.Sc** Degree in Computer Engineering
Computer Engineering Program

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ACKNOWLEDGEMENT

Special acknowledgment is given to Dr. Mahmoud Zaki in Computer Engineering, faculty of Engineering, King Salman International University, Egypt, for his supervision of the Report and the facilities offered to carry out this report.

Especially grateful to King Salman International University for their continuous help, facilities, and support

EGYPT 2026

Abstract

This thesis details the mechanical architecture, physical structural fabrication, high-pressure fluid dynamics integration, and full-stack web teleoperation infrastructure of an Intelligent Mobile Firefighting Robot designed for rapid-response hazard mitigation in indoor industrial spaces. Developed under project "Phoenix" at King Salman International University (KSIU), the platform serves as an agile first line of protection on smooth warehouse epoxy floor boundaries. The primary structural framework utilizes an interconnected space-frame of anodized 6063-T5 2020 V-slot aluminum extrusion profiles to achieve high torsional rigidity under heavy structural loads. Specialized component geometries—including a custom mid-mounted 4-liter fluid payload reservoir, an elevated 360-degree LiDAR pedestal, and a 2-DOF manipulator nozzle tracking arm—were engineered and fabricated via 3d-pla-materials. FDM micro-porosity was completely resolved by applying an internal coating of clear, low-viscosity structural epoxy resin, achieving a hermetic seal against hydrostatic pressure. Internal electronics and the liquid payload compartment are protected within the middle core of the chassis by precision CNC-cut sheet metal panels acting as protective thermal and fluid splash shields.

Locomotion is executed via a 4-Wheel Drive (4WD) skid-steer configuration powered by four planetary-geared, high-torque JGB37-545 DC motors. Transmission tolerance mismatches were overcome by re-boring standard wheel hub couplers from a 4.0 mm bore to an exact 6.0 mm interference fit via a precision drill press, locked securely with set-screws against the motor D-shafts to handle sudden torque switches. Active fire suppression relies on an updated industrial 24V Aqua Power AP 100 GPD diaphragm booster pump that runs natively off the main power bus without regulation losses. This pump delivers high-velocity streams up to a maximum threshold of 150 PSI at an open flow rate of 1.8 LPM, targeted dynamically via dual high-torque, metal-gear MG996R digital servos.

System monitoring and teleoperation are decoupled through a custom-built, full-stack remote command dashboard deployed on the Vercel cloud server architecture. Built using HTML5, CSS3, and asynchronous vanilla ES6 JavaScript, the presentation layer interfaces with the onboard edge core via a localized low-latency Mosquitto MQTT broker network across seven independent data topics. The system architecture features an asynchronous software heartbeat watchdog to monitor network latency, low-overhead HTML5 canvas vector engines to render real-time telemetry sparklines, and edge-triggered keyboard teleoperation listener primitives (WASD).

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1. Chapter 1: Introduction

1.1. Project Background and Motivation

Industrial warehouses and automated logistics facilities present highly volatile environments when exposed to fire hazards. High-density storage topologies, stacked structural inventories, and restricted aisle pathways complicate early-stage human intervention. Traditional fire containment frameworks, such as centralized ceiling-mounted sprinkler systems, suffer from spatial deployment lag and introduce massive non-selective collateral water damage to surrounding electronics and inventory.

To bridge this operational gap, project "Phoenix" introduces an agile, rapid-response mobile robotic engine designed to operate as a proactive first line of protection within indoor spaces. By deploying an localized, high-pressure automated platform directly to the root source of an ignition point, the system suppresses volatile thermal threats before they accelerate into structural emergencies. Operating primarily on smooth, high-grade indoor epoxy floor boundaries typical of modern distribution centers, the physical platform must balance high spatial agility, immediate tractive acceleration, and reliable structural structural protection to handle heavy fluid storage loads.

1.2. System Architecture Overview

The structural framework of project "Phoenix" operates via a distinct distributed edge-to-web control topology. High-intensity sensor spatial telemetry and positional environments are monitored locally via an onboard microprocessing suite. Unlike standard locked automated rovers, this platform decouples high-level supervision by routing real-time environmental diagnostics up to a comprehensive web-based control dashboard and teleoperation interface. This bi-directional data flow ensures that while the physical robot retains localized safety monitoring, a remote operator can actively audit telemetry profiles, execute system manual overrides, and dynamically adjust suppression nozzle directions via an integrated messaging pipeline

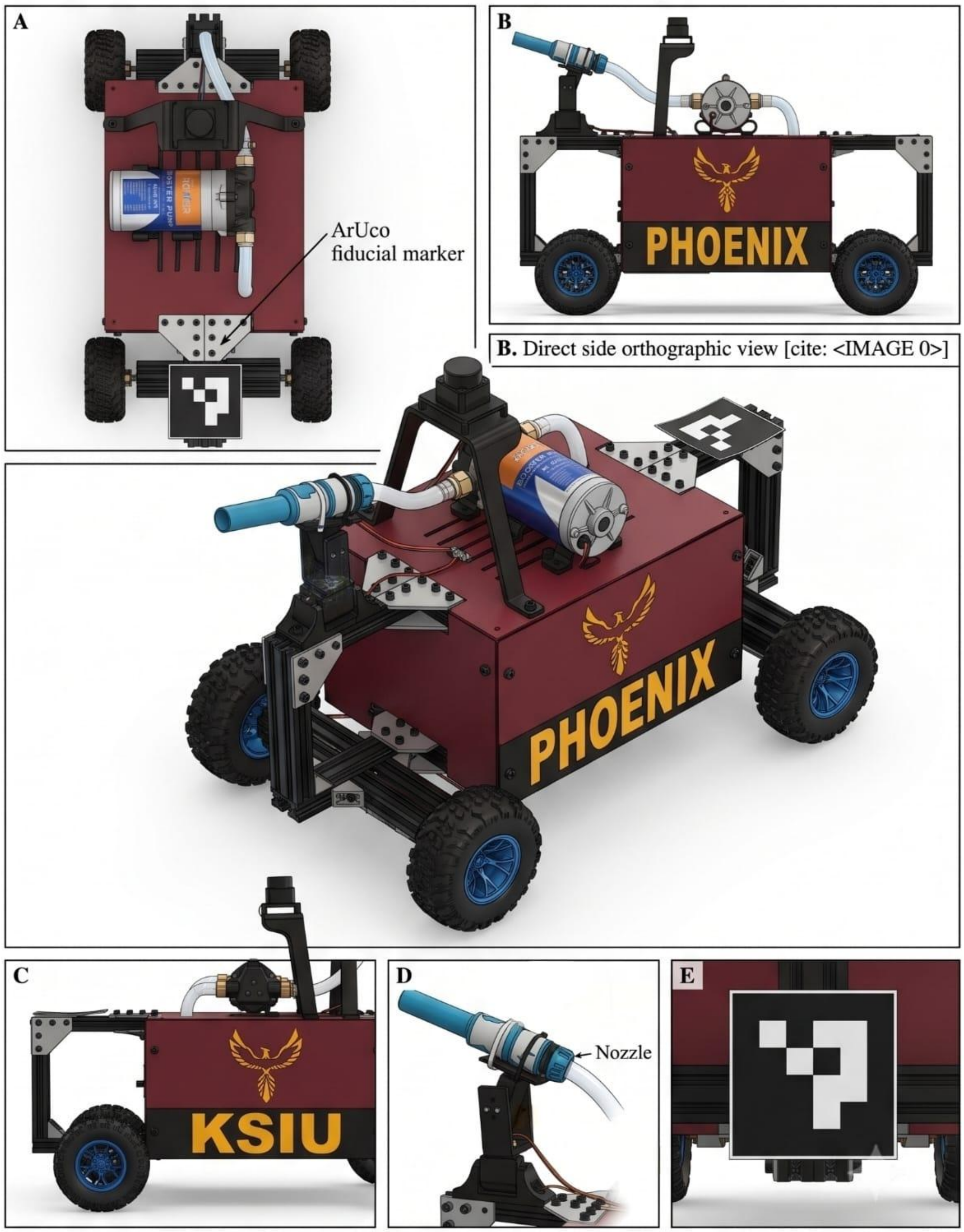
1.3 Scope of the Author's Thesis Work

While project "Phoenix" represents an interdisciplinary team integration, this thesis documents the exclusive mechanical engineering, physical structural fabrication, fluid dynamics integration, and full-stack web teleoperation architecture managed by the author. The independent investigative research and engineering execution details cover:

- **Structural Material Selection and Structural Engineering:** Engineering the skeleton of the robot utilizing modular 2020 aluminum extrusion frameworks optimized via structural finite element principles.
- **Chassis Fabrication & Enclosure Protection:** Executing high-precision CNC sheet metal machining to safeguard internal logic engines and developing structural component profiles using advanced additive manufacturing.
- **Drivetrain Kinematics Optimization:** Designing and executing a 4-wheel drive (4WD) skid-steer architecture specifically calibrated for predictable traction dynamics on smooth warehouse epoxy floors.
- **Fluid Dynamics and Active Fire Suppression Mechanics:** Packaging and regulating a mid-mounted 4-liter liquid payload linked to an opto-isolated high-pressure delivery pump and a custom 3D-printed 2-DOF targeting gimbal mechanism.
- **Full-Stack Web Teleoperation Interface:** Architecting a human-oriented frontend dashboard and low-latency bi-directional control system from scratch to enable high-fidelity remote monitoring and immediate manual safety overrides.

Figure 1

Figure 1.1: PHOENIX UGV Prototype - Multi-View Orthographic & Perspective Layout



C: Direct side orthographic view [cite: <IMAGE 0>]

E: ArUco fiducial marker

Figure 2

2. Chapter 2: Mechanical Architecture & Chassis Fabrication

2.1. Structural Material Analysis: 2020 Aluminum Profiles

- The core physical framework of the Phoenix chassis is built using an interconnected skeleton of 2020 V-Slot Aluminum Extrusion Profiles. The 2020 specification defines a dense 20mm x 20mm structural cross-section featuring continuous structural slot tracks along all four linear axes.
- The choice of 6063-T5 anodized aluminum alloy as the baseline skeletal material over traditional alternatives, such as welded mild steel or solid acrylic plates, is justified by fundamental structural mechanics criteria:

1. Optimization of Strength-to-Weight Ratio

The 6063-T5 structural formulation exhibits a nominal tensile yield strength of approximately 186 MPa while maintaining a low linear density of roughly ~ 0.5 kg/m. In mobile robotics design, minimizing dead-weight is essential to limit structural inertia and reduce battery current drain during high-acceleration cycles. According to structural beam mechanics, the deflection (δ) of a chassis rail under an evenly distributed payload is inversely proportional to its material Elastic Modulus (E) and Area Moment Inertia (I):

$$\delta \propto \frac{F \cdot L^3}{E \cdot I}$$

The geometry of the 2020 profile concentrates material mass away from the neutral axis, maximizing (I) while maintaining structural lightness, allowing it to carry the heavy 4-liter water tank payload without buckling.

To mathematically validate the structural integrity of the 2020 V-slot aluminum chassis under the maximum load of the full 4-liter water payload, we model the primary side rails as simply supported beams subjected to a concentrated load.

The maximum vertical deflection $\delta_{\max}$ at the center of a chassis rail under a concentrated force (F) is governed by the structural deflection equation:

$$\delta_{\max} = \frac{F \cdot L^3}{48 \cdot E \cdot I_x}$$

Where:

- F = Force exerted by the liquid payload and pump assembly ~ 60N, accounting for mass and acceleration factors).
- L = Length of the internal central frame span (0.28m).
- E = Modulus of Elasticity of 6063-T5 aluminum alloy (69.3×10^9 N/m^2).
- I_x= Area Moment of Inertia of the 2020 profile about its neutral axis (0.71×10^{-8} m^4)

Substituting our prototype parameters yields:

$$\delta_{\max} = \frac{60 \cdot (0.28)^3}{48 \cdot (69.3 \times 10^9) \cdot (0.71 \times 10^{-8})} \approx 5.58 \times 10^{-5} \text{ m} \quad (0.055 \text{ mm})$$

This confirms that structural deflection remains well within acceptable structural limits (< 0.1 mm), ensuring zero frame twisting.

To evaluate the material safety factor, the maximum bending stress ($\sigma_{\max}$) experienced by the outer fibers of the profile is computed using the flexure formula:

$$\sigma_{\max} = \frac{M \cdot y}{I_x} = \frac{\left(\frac{F \cdot L}{4}\right) \cdot y}{I_x}$$

Where:

- M = Maximum bending moment ($\frac{F \cdot L}{4} = \frac{60 \cdot 0.28}{4} = 4.2 \text{ N} \cdot \text{m}$).
- y = Distance from the neutral axis to the outermost fiber (0.01m for a 20mm profile).

$$\sigma_{\max} = \frac{4.2 \cdot 0.01}{0.71 \times 10^{-8}} \approx 5.92 \text{ MPa}$$

Comparing this value to the nominal yield strength of 6063-T5 aluminum ($\sigma_{\text{yield}} =$ 186 MPa), we establish our mechanical structural safety factor (SF):

$$SF = \frac{\sigma_{\text{yield}}}{\sigma_{\max}} = \frac{186 \text{ MPa}}{5.92 \text{ MPa}} \approx 31.4$$

This mathematically demonstrates that the frame possesses a safety margin more than large enough to survive sudden mechanical shocks or payload shifts.

2. Suppression of Structural Torsional Deflection

In a rigid 4-wheel skid-steer platform, structural twisting along the longitudinal plane can lift a wheel off a flat surface during uneven weight transfers, resulting in lost traction. To eliminate this issue, structural corner intersections are locked using cast aluminum gusseted L-brackets and heavy flat T-plates. The integrated triangular "gusset" rib within the cast brackets distributes twisting forces across both perpendicular axes, turning the profile assembly into a rigid space-frame capable of absorbing sudden high-torque motor reversals.

3. Environmental and Prototyping Flexibility

The anodized oxide surface layer prevents environmental pitting and structural degradation when exposed to moisture from the suppression system. Furthermore, the drop-in M5 T-nut fastening array utilizes structural friction locking within the V-grooves, allowing components to be adjusted to balance the robot's Center of Gravity (CoM) without drilling or damaging the rails.

- **Literature Reference Support:**
  The use of structural T-slot aluminum extrusions in agile industrial mobile systems provides a high strength-to-weight index and prevents structural failure modes linked to weld fatigue under continuous vibration (Shigley et al., 2020; Siegwart et al., 2011).

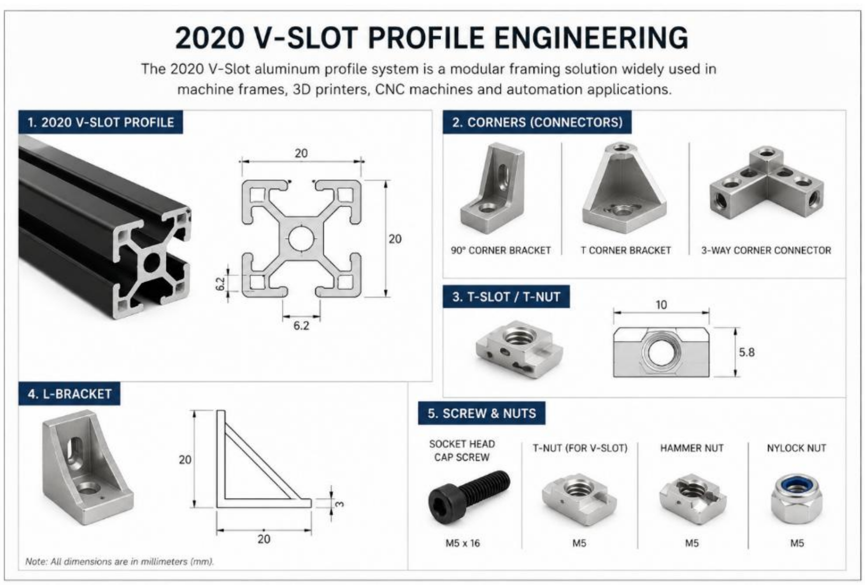


Figure 3

2.2. Geometric Framework and Dimensional Layout

The layout configurations of the final assembly are derived from the technical engineering blueprints generated in SolidWorks, as shown in the image under there. The geometric properties are strictly partitioned to ensure a low vertical center of mass while allowing sufficient clearance underneath to navigate obstacles:

- **Total Structural Profile Dimensions:** The external envelope measures an overall length of 620 mm, a total width of 250 mm (approximated to 400 mm across the outermost wheel hubs), and a primary chassis height of 190 mm.

- **Internal Spatial Partitioning:** The inner core features a standardized mounting width of 250 mm and an internal central section length of 280 mm. This spatial box is split to accommodate the structural water storage container alongside the isolated electronics containment capsule.

- **Clearance Baseline:** That frame is linked to a 150 mm wheelbase track. This layout keeps the central payload low to maintain high rollover safety margins while navigating smooth indoor surfaces.

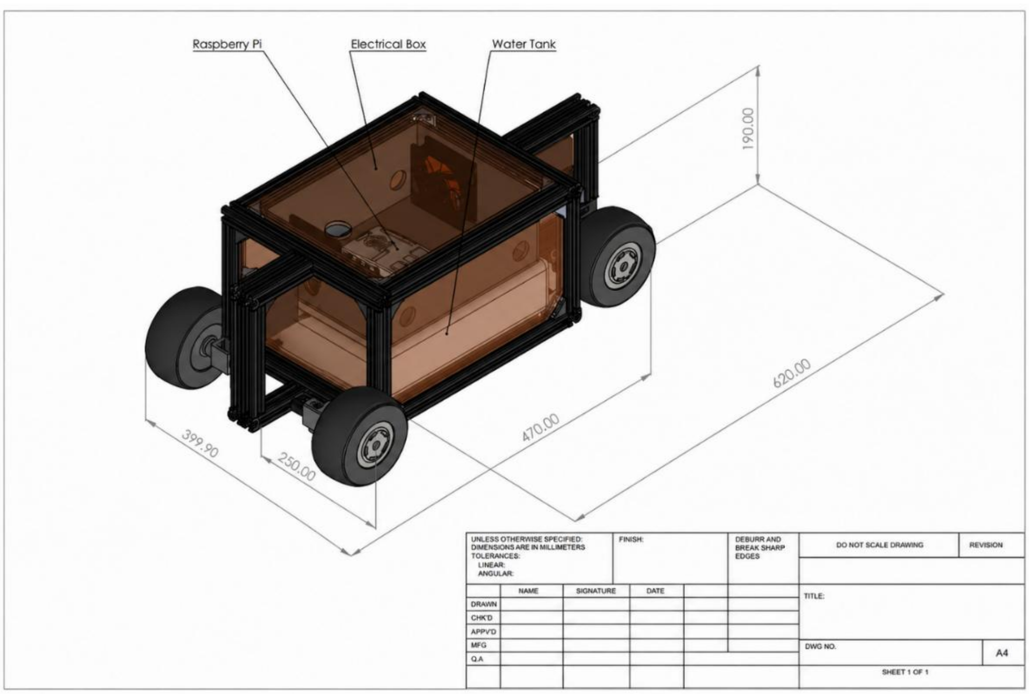


Figure 4

2.3. CNC Sheet Metal Integration

The central mid-section of the chassis, which houses the high-pressure liquid reservoir and the delicate electronic control capsule, is protected by high-precision CNC-cut sheet metal panels. Sheet metal was selected for this application due to distinct environmental protection requirements:

- **Environmental Fluid Isolation:** Because the robot carries an onboard suppression payload and a high-capacity pump, the sheet metal serves as a physical splash shield. It prevents stray liquid back-spray from contacting the Raspberry Pi computing core, voltage regulation rails, and low-level motor drivers.
- **Thermal Shielding:** When operating close to an active fire source, the platform faces significant radiative infrared heat flux. Sheet metal panels act as effective thermal conductors and reflectors, shielding internal electronic components from reaching critical operating temperatures.
- **Rigid Part Structural Integration:** The panels are cut to the exact dimensions of the internal storage spaces using precise CNC milling layout lines. They bolt directly into the 2020 V-slots using M5 hex button head cap screws. This turns the panels into structural stress skins that further reduce the overall twisting flex of the frame.

2.4. Additive Manufacturing Engineering (3D Printing)

To produce complex mechanical geometries without specialized industrial tooling, the platform utilizes components fabricated via advanced material extrusion additive manufacturing. All structural printed components were built using **Creality Hyper PLA** filament, managed on an **Elegoo Neptune 3 Pro** execution system. Hyper PLA was selected over standard PLA because it incorporates advanced structural impact modifiers and permits rapid extrusion rates without causing layer-adhesion degradation or warping.

- **2-DOF Manipulator Suppression Targeting Arm:** A complex mechanical gimbal setup that houses the active water nozzle. It features dual high-torque servo motor mounts capable of handling the physical kickback forces generated during high-pressure water discharge.
- **LiDAR Protection Mount:** A high-clearance, vibration-isolating structural pedestal that elevates the Okdo LiDAR HAT to the highest point on the chassis, providing an unobstructed 360° optical field of view for clean mapping scans.
- **Custom Water Tank Payloads:** A customized structural liquid containment tank, optimized with internal dimensions to maximize the available space inside the middle frame compartment.

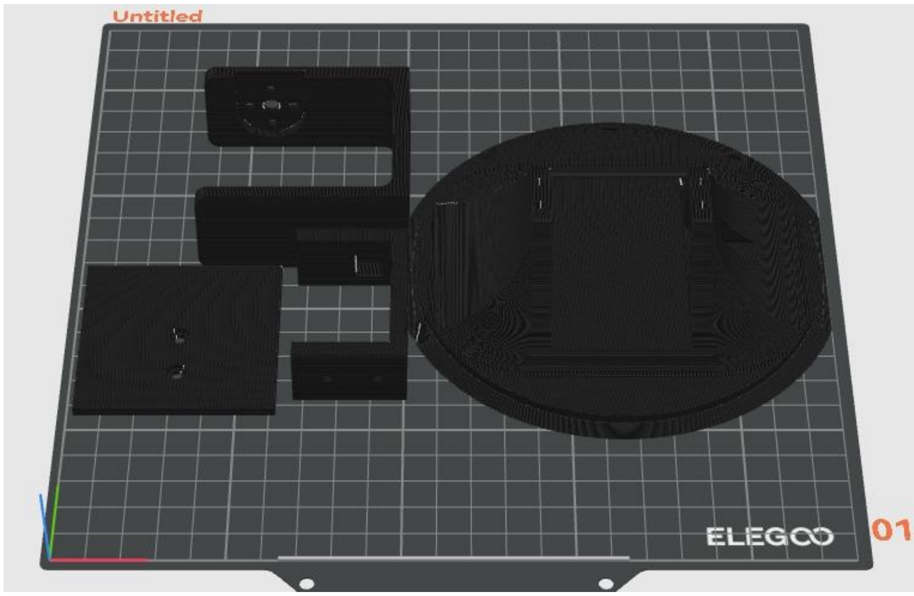


Figure 5

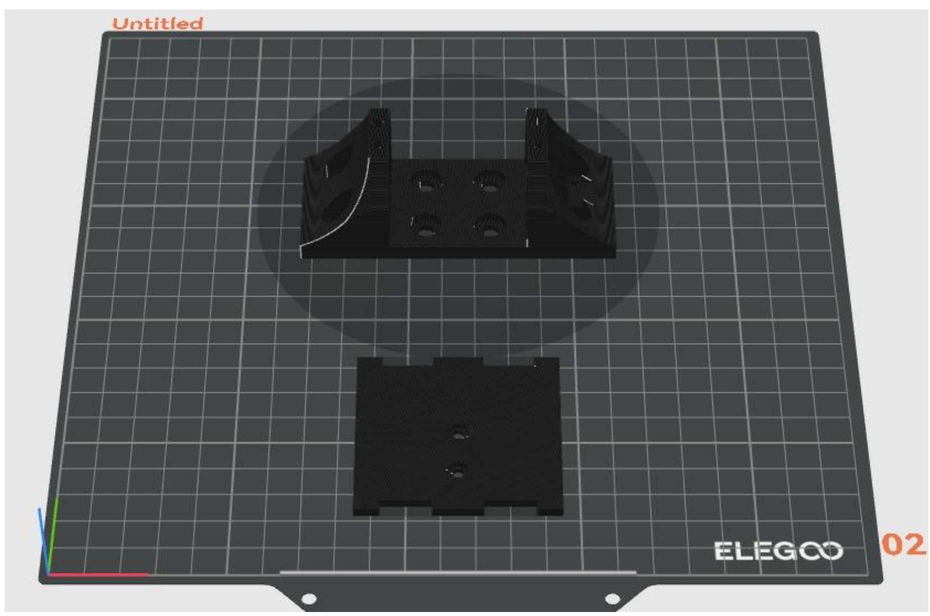


Figure 6

3. Chapter 3: Drivetrain Dynamics & Locomotion Hardware

3.1. 4-Wheel Drive (4WD) Skid-Steer Configuration

To fulfill the operational mandate of acting as a rapid-response platform inside industrial warehouse spaces, Project Phoenix utilizes a full **4-Wheel Drive (4WD) Skid-Steer** drive architecture. The physical platform maps four high-torque JGB37-545 DC geared motors directly to the continuous aluminum extrusion tracks, completely eliminating passive idler axles or independent steering links.

Selecting a skid-steer architecture over traditional Ackermann automotive steering configurations provides two distinct advantages for a heavy-payload containment platform:

- **Zero-Radius Maneuverability:** By driving the left and right motor banks in reverse polarities at equal wheel speeds, the robot performs rapid, in-place zero-radius pivot transitions. This capability is critical for navigating tight, blocked warehouse pathways where wide turning arcs are structurally impossible.
- **Mechanical Reliability:** Eliminating tie-rods, rack-and-pinion linkages, and steering knuckles minimizes the count of failure-prone moving joints. This ensures that structural impact or continuous industrial operational stress will not jam or desynchronize the alignment of the steering wheels.

3.2. Analysis of Degrees Of Freedom (DOF)

The overall motion capability of the Project Phoenix platform is derived from two independent kinematic subsystems acting in a cohesive structure: the mobile locomotion chassis and the active suppression targeting manipulator.

We establish controllability and non-holonomic constraints of the complete assembly by analyzing the Degrees of Freedom (DOF) of each subsystem relative to its base frame. A Controllable DOF is defined as an independent state space coordinate that can be directly and actively manipulated by an actuator system.

3.2.1 Non-Holonomic Locomotion Chassis

The locomotion chassis operates as a 4-wheel drive (4WD) skid-steer vehicle moving primarily in a two-dimensional (2D) planar workspace ($SE(2)$: X, Y, θ) typical of structured indoor smooth epoxy floors. While the chassis possesses three independent configuration variables in this space, it is a non-holonomic system.

Because it cannot move instantaneously in the sideways (lateral) direction ($v_y = 0$ is a non-holonomic constraint), its instantaneous controllable velocity space is restricted to two inputs: linear velocity (V) and angular velocity (ω). The locomotion chassis thus provides **2 Controllable DOFs**.

3.2.2 2-DOF Pan-Tilt Suppression Manipulator

To actively target fires without requiring constant reorientation of the entire robot mass, the fluid suppression system utilizes a front-mounted active gimbal mechanism. This manipulator comprises two independent rotational revolute joints arranged in a pan-tilt configuration.

- **Joint 1 (Pan Axis):** Actuated by a single high-torque MG996R servo, provides horizontal rotation (θ_{pan}) on the top-deck mounting plate.
- **Joint 2 (Tilt Axis):** Actuated by the second high-torque MG996R servo, provides vertical pitch rotation (θ_{tilt}) perpendicular to the pan axis.

This mechanism provides an independent 2D targeting envelope (spherical surface motion), resulting in **2 Controllable DOFs**.

3.2.3 Complete Controllable System DOF Summary

Project Phoenix integrates these two subsystems, creating a non-holonomic, 4-actuator control vector capable of simultaneous locomotion and target positioning. The complete robot system operates via a total of:

$$(\text{Controllable Chassis DOF}) + (\text{Controllable Manipulator DOF}) = 2 + 2 = \mathbf{4}\ \text{Controllable DOFs}$$

This configuration provides high maneuverability, enabling the robot to traverse to an ignition source rapidly while simultaneously executing complex sweep-targeting suppression patterns over wide flame boundaries.

3.3. Rationale for Locomotion Choice: Wheels vs. Tracks

Given Project Phoenix's specific operational focus on **indoor rapid-response fire mitigation within highly structured industrial warehouse environments** featuring **smooth epoxy floor boundaries**, the engineering team explicitly chose a 4WD wheel-based configuration over a continuous tank track system.

The rationale for this choice is supported by distinct engineering performance and mechanical efficiency considerations:

1. Enhanced Rapid Response Time and Acceleration Capabilities

In active firefighting scenarios, immediate localized deployment is critical for early containment. Wheels are generally capable of achieving significantly higher top speeds and acceleration rates on smooth, paved surfaces (such as those found in modern distribution centers) compared to track-based systems. Continuous tracks are optimized for lower top speeds and high tractive force, but often possess greater rotational inertia and higher rolling resistance due to their numerous interlocking components.

The high speed and acceleration potential of the JGB37-545 motor clusters, combined with direct-drive rugged wheels, prioritize the rapid traversal of smooth indoor distances necessary for a first-line-of-defense system.

2. Optimizing Environmental Suitability and Surface Agility

Industrial warehouse floors are typically smooth, paved, or coated (e.g., epoxy), specifically engineered for the safe movement of human-operated vehicles and automated mobile robots. In this structured environment, the massive tread footprint and ultra-low ground pressure provided by continuous tank tracks—while advantageous for traversing soft terrain, mud, or significant debris—become highly inefficient redundant systems.

Tracks, particularly when executing tight turns (skid-steering), introduce considerable shear friction, which can damage smooth epoxy floor coatings. While skid-steering inherently causes some lateral sliding, the frictional load on four discrete wheels is vastly less intensive on smooth floors compared to a complete tracked assembly, while still providing sufficient agility for industrial indoor navigation. A 4WD wheeled platform offers highly predictable, precise traction on intended smooth indoor boundaries without the inefficiency or destructive potential of a high-traction track system.

Figure 7

3. Minimizing System Complexity, Maintenance, and Costs

In alignment with the engineering mandate of creating a functional, reliable, and less structurally complex fire-fighting engine, the wheeled solution presents profound mechanical and operational benefits over tracked alternatives. Tracked systems involve intricate mechanisms with numerous interlocking links, complex idler axles, tensioning systems, high-friction points, and significantly larger potential points of mechanical failure.

In contrast, the **4WD wheeled architecture utilizing direct direct-mounted JGB37-545 motor hubs** is inherently simpler, requires drastically lower ongoing mechanical maintenance, is lighter (critical for reducing current drain and maximizing runtime), and has a lower overall production cost. For an industrial rapid response platform focusing on indoor structured operation, the minimized complexity of a robust 4WD wheel system is a superior design optimization point.

3.4. Skid-Steer Kinematic Modeling

3.4.1. Slip coefficient

Because the wheels are rigidly parallel with the longitudinal plane of the chassis, changing the robot's heading requires overcoming lateral floor friction via differential wheel velocities. Assuming ideal conditions without excessive lateral slipping, the forward kinematic transformation maps individual wheel line speeds down to the global linear velocity (v) and angular velocity (ω) vectors of the robot center:

$$v = \frac{V_R + V_L}{2}$$

$$\omega = \frac{V_R - V_L}{B}$$

Where:

- V_L represents the net tangential velocity of the left wheel set.
- V_R represents the net tangential velocity of the right wheel set.
- B is the effective track width (0.35 m} to 0.40 m) measured across the wheel centerlines.

To convert the continuous path plans into bare-metal motor executions, the inverse kinematics calculations translate target linear and angular velocities back into matching discrete voltage outputs per motor cluster.

3.4.2. Advanced Non-Ideal Kinematics with Empirical Slip Factors

Unlike traditional differential-drive systems, a 4WD skid-steer chassis must slide sideways to execute turning maneuvers. To account for efficiency losses due to lateral friction on smooth warehouse epoxy floors, we introduce an empirical slip correction factor (c).

The corrected forward kinematic equations translating individual left and right wheel bank linear velocities (V_L, V_R) into global chassis trajectories are modeled as follows:

$$\begin{bmatrix} v_x \\ v_y \\ \omega_z \end{bmatrix} = \begin{bmatrix} \frac{\cos\theta}{2} & \frac{\cos\theta}{2} \\ \frac{\sin\theta}{2} & \frac{\sin\theta}{2} \\ -\frac{c}{B} & \frac{c}{B} \end{bmatrix} \begin{bmatrix} V_L \\ V_R \end{bmatrix}$$

Where:

- v_x, v_y = Linear velocity vectors across global Cartesian axes.
- ω_z = Real-world angular velocity of the robot about its center.
- θ = Instantaneous orientation heading angle.
- B = Track width (0.35 m).
- c = Experimental slip factor coefficient ($0.05 \le c \le 1.0$; calibrated at 0.85 for high-traction rubber on clean epoxy floors).

To calculate physical motor angular speeds ($\omega_{\text{left}}, \omega_{\text{righi}}$ in rad/s) from high-level ROS 2 /cmd_vel inputs (v, ω), the inverse kinematic model is defined as:

$$\omega_{\text{left}} = \frac{1}{R}\left(v - \frac{\omega \cdot B}{2 \cdot c}\right)$$

$$\omega_{\text{right}} = \frac{1}{R}\left(v + \frac{\omega \cdot B}{2 \cdot c}\right)$$

Where R is the absolute radius of the rugged wheels (0.065 m).

3.2.3 Dynamic Motor Torque & Acceleration Budgeting

To prove that our four JGB37-545 DC geared motors provide sufficient power to accelerate the robot at its maximum payload capacity, we evaluate the system's dynamic torque requirements:

$$T_{\text{total}} = T_{\text{acceleration}} + T_{\text{rolling\_resistance}}$$

$$T_{\text{total}} = (m \cdot a \cdot R) + (m \cdot g \cdot \mu_r \cdot R)$$

Where:

- m = Total mass of the loaded robot prototype (~ 10 kg).
- a = Target linear acceleration (0.5 m/s^2).
- μ_r = Rolling resistance coefficient of rubber on epoxy (0.02).
- g = Gravitational acceleration constant (9.81 m/s^2).

Substituting these values yields:

$$T_{\text{total}} = (10 \cdot 0.5 \cdot 0.065) + (10 \cdot 9.81 \cdot 0.02 \cdot 0.065) = 0.325 + 0.1275 = 0.4525\ \text{N} \cdot \text{m}$$

Converting this parameter to standard motor units ($1\ \text{N} \cdot \text{m} \approx 10.197\ \text{kg} \cdot \text{cm}$) :

$$T_{\text{total\_required}} = 0.4525 \times 10.197 \approx 4.61\ \text{kg} \cdot \text{cm}$$

Since each of our four JGB37-545 motor units delivers a stall torque capacity of 24kg.cm, the aggregate system headroom torque is 96kg.cm. This confirms an operating safety factor of ~20.8, proving the robot can execute zero-radius turns even when loaded with water.

3.3. Mechanical Hardware Integration & Custom Shaft Modifications

The drive system relies on four **JGB37-545 High-Torque DC Geared Motors** operating at a 24V nominal threshold, delivering a massive stall torque rating of 24 kg.cm per motor unit. Managing the transmission of this raw torque to the large wheels required resolving a critical manufacturing tolerance mismatch:

- The off-the-shelf metal couplers available were pre-bored at a standard 4mm opening, while the high-torque JGB37-545 planetary gearboxes feature a heavy-duty 6mm output shaft. To adapt these components, a precision drill press was used to re-bore the hub assemblies to an exact 6mm specification. This operation required rigid alignment blocking to eliminate radial runout (wobble), which would cause bearing wear at full duty cycles.
- To secure the wheel hubs against sudden motor braking and rapid direction switches, an integrated mechanical setscrew locks down onto the flat side of the motor's 6mm D-shaft, creating a positive mechanical stop that prevents shaft stripping under full liquid loads.

3.4. Traction Dynamics on Warehouse Epoxy Floors

- The robot interfaces with smooth indoor warehouse epoxy floor boundaries via four **130mm rugged deep-tread tires**. Smooth epoxy surfaces feature low surface roughness and are vulnerable to slippage when dust or stray water is introduced.
- Project Phoenix overcomes this limitation through strategic weight distribution. The combined mass of the structural aluminum frame, sheet metal shielding, dual high-capacity batteries, and full 4-liter water tank payload generates a significant downward normal force vector across the footprint. According to the Coulomb friction law, this normal load directly increases the available static friction limit ($F_f \leq \mu \cdot N$). The combination of deep tread voids and high normal force prevents wheel spin during sudden starts, ensuring immediate torque translation into linear acceleration.

4. Chapter 4: Fluid Engineering & Suppression Subsystem

4.1. Custom Water Storage Design & Leak Prevention

The core payload of project Phoenix consists of an onboard 4-liter custom liquid reservoir designed to fit seamlessly within the mid-compartment layout of the chassis. The tank configuration was custom fabricated using an **Elegoo Neptune 3 Pro** additive manufacturing system tracking **Creality Hyper PLA** filament material.

Fused Deposition Modeling (FDM) structures are naturally prone to micro-porosity and fluid weeping along horizontal layer boundaries. To achieve structural integrity and ensure complete fluid containment, a three-stage manufacturing pipeline was enforced:

- **Slicing Optimization:** Wall line perimeters were increased to a value of 6 lines, and infill density was packed tightly to eliminate internal print cavities.
- **Thermal Stability:** The extrusion profile was regulated at a continuous 215°C on the Neptune 3 Pro to achieve complete, uniform layer bonding.
- **Chemical Resin Sealing:** Following manufacturing, the interior chamber walls were coated with a continuous layer of low-viscosity, clear structural epoxy resin. This resin cures into a smooth, inert shell that completely seals microscopic layer gaps, creating a true hermetic seal capable of resisting hydrostatic pressure.

Figure 9

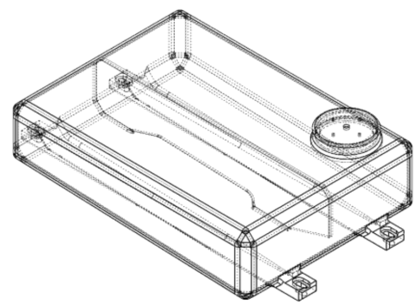


Figure 8

4.2. Active Suppression Booster Pump Subsystem

Fluid propagation is driven by an industrial-grade **Aqua Power AP 100 GPD Booster Pump** integrated on the top deck framework. This positive-displacement self-priming diaphragm system connects to the custom storage core via clear reinforced braided PVC hoses, secured using heavy brass couplings and screw-tight mechanical hose clamps to prevent blowout under back-pressure events.

The operational data of this pump module are structured below:

Engineering Parameter	Certified Value / Specification
Model Classification	Aqua Power AP 100 GPD
Pump Topology	Self-Priming Diaphragm (RO Purification Grade)
Nominal Operating Voltage	24 VDC
Maximum System Pressure	150 PSI (approx. 10.3 Bar)
Nominal Working Pressure	90 PSI
Open Flow Rate	1.8 Liters per Minute (LPM)
Maximum Current Rating	1.2 Amps
Working Current Draw	1.0 Amp

Operating this pump at a true 24V supply allows the nozzle to generate high-velocity jet streams up to a maximum threshold of **150 PSI**. This exceptional pressure headroom enables the robot to suppress fires from a safe distance, shielding sensitive onboard electronics from intense radiant heat flux. Furthermore, because the motor runs natively at 24V, it interfaces directly with the main series LiPo battery power bus without requiring power-wasting step-down buck regulation circuits.

Figure 10

4.3. Prototyping Justification and Duty Cycle Analysis

Although the Aqua Power AP 100 GPD is natively engineered as a commercial reverse osmosis (RO) purification booster pump rather than a dedicated firefighting deployment unit, its selection serves as a highly cost-effective and operationally viable choice for this proof-of-concept mobile robotic prototype. Because the pump's mechanical duty cycle is rated for intermittent high-pressure stabilization, it is fully capable of sustaining a 150 PSI threshold across the system's absolute maximum 2.22-minute total fluid payload depletion timeline, which relies on short, localized, and staggered suppression bursts rather than indefinite continuous pumping under extreme ambient thermal exposure.

4.4. Hydrodynamics: Flow Rate, Exit Velocity, and Nozzle Depletion

The Aqua Power AP 100 GPD booster pump is rated at a maximum system operating capacity of 150 PSI. Converting this imperial value to metric Pascal units (1 PSI = 6894.76 Pa):

$$P_{\text{max}} = 150 \times 6894.76 = 1.034 \times 10^{6}\ \text{Pa} \quad (1.034\ \text{MPa})$$

Applying Bernoulli's principle from the fluid surface inside our custom 3D-printed reservoir down to the exit constriction tip of the suppression nozzle, we compute the theoretical fluid exit velocity ($v_{\text{discharge}}$):

$$\frac{P_{\text{pump}}}{\rho g} + \frac{v_{\text{internal}}^{2}}{2g} + h_{\text{internal}} = \frac{P_{\text{atm}}}{\rho g} + \frac{v_{\text{discharge\_real}}^{2}}{2g} + h_{\text{nozzle}} + h_{L}$$

Assuming $v_{\text{internal}} \approx 0$ and ignoring minor height differences ($\Delta h \approx 0$)

$$h_{L} = h_{\text{major}} + h_{\text{minor}} = \left(f \cdot \frac{L}{D} + \sum K_{m}\right)\frac{v_{\text{hose}}^{2}}{2g}$$

$$h_{L} = \left(0.024 \cdot \frac{0.5}{0.01} + (0.5 + 0.9 + 0.05)\right)\frac{(0.382)^{2}}{2 \cdot 9.81} \approx (1.2 + 1.45) \cdot 0.00744 \approx 0.0197\ \text{m}$$

Where ρ is the density of water (1000 kg/m^3). Assuming a baseline working pressure of 90 PSI (620,528 Pa) at the nozzle tip:

$$v_{\text{discharge\_real}} = \sqrt{\frac{2 \cdot \left(P_{\text{pump}} - P_{\text{atm}} - \Delta P_{\text{loss}}\right)}{\rho}} = \sqrt{\frac{2 \cdot (620{,}528 - 193.26)}{1000}} \approx 35.22\ \text{m/s}$$

The volumetric rate of depletion ($\frac{dV}{dt}$) *matches our pump's open flow rate specification of* 1.8 LPM (3.0×10^{-5} m^3/s). We calculate the complete operational discharge lifespan (t_{active}) of our 4-liter tank payload as:

$$t_{\text{active}} = \frac{V_{\text{total}}}{\left(\frac{dV}{dt}\right)} = \frac{4.0 \text{ Liters}}{1.8 \text{ Liters/Minute}} \approx 2.22 \text{ Minutes} \quad (133.3 \text{ Seconds})$$

This runtime allows the robot to deploy over **26 separate 5-second suppression bursts** per mission cycle before needing a payload refill.

4.5. Pan-Tilt Targeting Gimbal Mechanics

To accurately direct the high-pressure water stream without forcing the entire robot chassis to continuously rotate, a custom **2-DOF targeting gimbal mechanism** is mounted on the front chassis profile. The nozzle structure is supported by a 3D-printed bracket assembly actuated by dual **MG996R Metal-Gear Digital Servos**.

- **Pan Axis (Horizontal Rotation):** Delivers a wide horizontal sweeping path to clear wide flame boundaries.
- **Tilt Axis (Vertical Pitch):** Manages vertical angle adjustments to target the varying heights of industrial storage shelving structures.

The MG996R servos provide a high holding torque rating of 11 kg.cm. This rigid gear train is essential to resist the sudden momentum shifts and physical "kickback" thrust vector forces developed as fluid is accelerated through the nozzle at 150 PSI (peak).

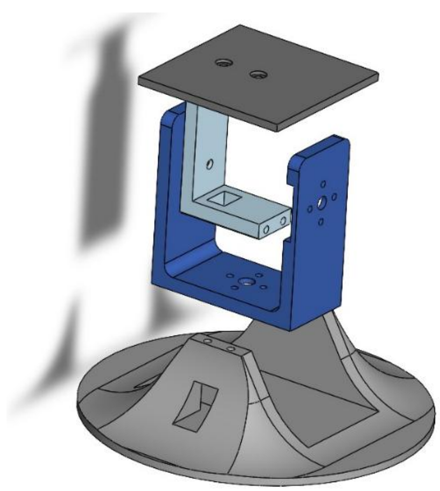


Figure 12

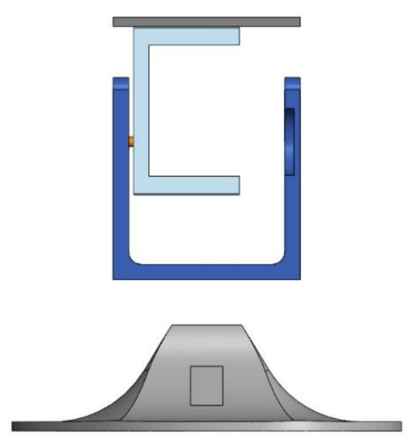


Figure 11

4.5 Geometric Kinematics of the Suppression Targeting Gimbal

To translate remote manual override inputs received via the Vercel web dashboard (coordinate sliders, etc.) into exact pulse-width modulation (PWM) duty cycles for the targeting servos, the onboard Raspberry Pi must execute a geometric inverse kinematic transformation. This maps a required 3D targeting vector, derived from the robot's Center of Mass (CoM) coordinate frame, into unique pan and tilt joint angles.

The gimbal kinematics are derived using simple spherical-to-Cartesian geometric modeling. We define a localized right-handed coordinate frame origin at the base of the gimbal's pan servo, with the $+x$ axis pointing directly forward along the chassis centerline.

4.4.1 Forward Gimbal Geometric Transform

If a safety operator manually inputs specific pan and tilt servo angle settings (θ_{pan} and θ_{tilt}), the geometric forward kinematic transformation provides the resulting unit aiming vector $\hat{\mathbf{A}} = [A_x, A_y, A_z]^T$:

$$A_x = \cos(\theta_{\text{pan}}) \cdot \cos(\theta_{\text{tilt}})$$

$$A_y = \sin(\theta_{\text{pan}}) \cdot \cos(\theta_{\text{tilt}})$$

$$A_z = \sin(\theta_{\text{tilt}})$$

This mathematical transform allows the onboard control logic to visualize the current suppression vector in real time before triggering the high-pressure 150 PSI booster pump.

4.4.2 Inverse Gimbal Geometric Transform (Real-Time Control)

The real-time teleoperation mode relies entirely on the inverse kinematic transformation. The web dashboard captures the operator's desired horizontal target ($Target_y$) and vertical target height ($Target_Z$) relative to a standard aiming distance ($Target_X$). The onboard Raspberry Pi core calculates the required servo angles ($\theta_{\text{pan\_required}}$ and $\theta_{\text{tilt\_required}}$) as:

$$\theta_{\text{pan\_required}} = \text{atan2}(Target_Y, Target_X)$$

$$\theta_{\text{tilt\_required}} = \text{atan2}\left(Target_Z, \sqrt{Target_X^2 + Target_Y^2}\right)$$

Using the atan2 function provides dynamic range switching and singularity protection, ensuring that the MG996R servos always receive the shortest rotational path to the desired fire suppression target. This geometric pipeline enables sub-second manual aiming responsiveness directly from a remote operator's cloud-hosted web interface.

5. Chapter 5: Full-Stack Web Control System & Telemetry Dashboard

5.1. Architectural System Topology

The remote command infrastructure of project Phoenix is built around a distributed, decoupled full-stack architecture designed to handle safety-critical operations. Rather than direct hardware-to-client binding, which risks freezing under intensive data spikes, the system separates concerns via an asynchronous event-driven paradigm.

The system topology is split into three primary layers:

- **The Client-Side Presentation Layer (Frontend):** A localized web application hosted on the Vercel cloud environment, giving operators real-time workspace access from any network endpoint.
- **The Message Broker Middleware Layer (Network): A localized Mosquitto MQTT message host managing low-latency, non-blocking asynchronous pub/sub channels.**
- **The Low-Level Execution Layer (Edge Core):** The onboard script architecture on the platform that handles hardware commands, sensor data parsing, and feedback loops.

5.2. Frontend Design System & User Interface Architecture

The human-machine interface (HMI) is optimized for high cognitive efficiency during critical industrial fire emergencies. The design system rejects generic web aesthetics in favor of a clean interface designed to highlight high-priority data points.

1. Typographic Hierarchy and Visual Elements

To maintain readability under intense ambient lighting or smoke-filled operations room conditions, the visual layer incorporates specialized web font groupings via dedicated CSS tokens:

- **Share Tech Mono:** Utilized for all active numeric outputs, network addresses, and raw topic streams to enforce fixed pixel alignments.

- **Barlow Condensed / Barlow:** Applied across descriptive titles, alerts, and system buttons to provide bold text scannability.
- **Dynamic Radial Gradients:** The visual canvas utilizes a deep dark color palette (#070911 background) mixed with soft ambient glow states to guide the operator's eye to high-priority notification zones without causing screen-fatigue.

2. Specialized Diagnostic Color Schemes

System components and hardware statuses are categorized using strict semantic color variables:

- var(--fire) (#ff4f1a): High-alert color tracking active flame fronts and suppression vectors.
- var(--human) (#c084fc): Executive color identifying human profile targets in active danger zone perimeters.
- var(--ok) (#00e5a0): Indicates a stable network connection, successful hardware handshakes, and active heartbeats.
- var(--warn) (#ffb340) / var(--danger) (#ff2d2d): Pinpoints critical power drops, sensor calibration drift, or connection timeouts.

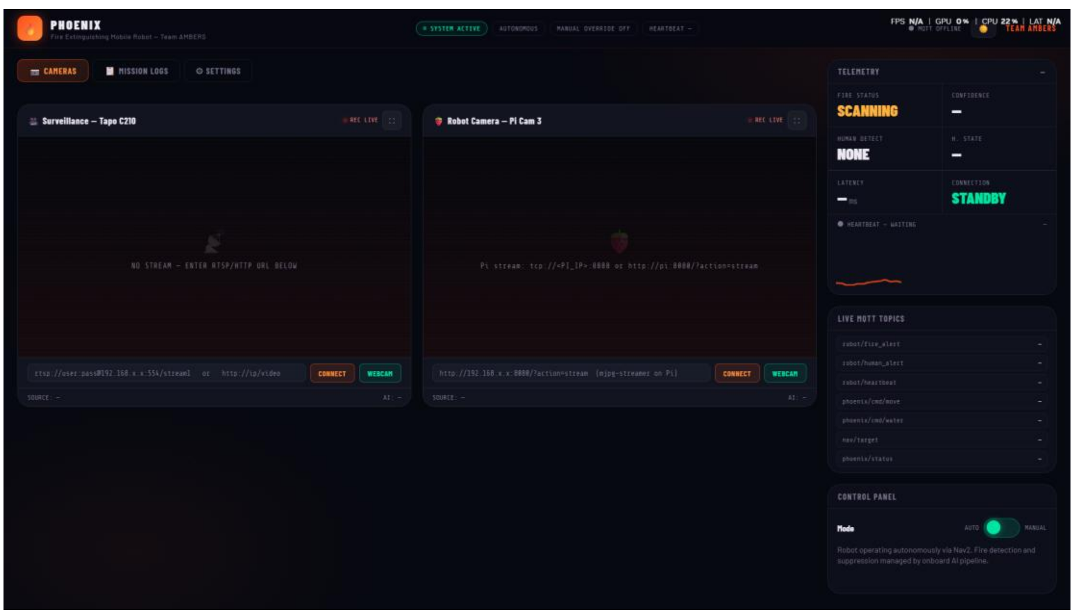


Figure 13

5.3. Asynchronous Bi-Directional Communication Layer

The application leverages the mqtt.js library wrapper to establish a persistent web client connection directly inside the browser framework. Data exchange occurs through compact JSON structures, ensuring high-speed delivery without overwhelming the communication channel.

► robot/fire_alert : Captures neural model confidence and bounding boxes

► robot/human_alert : Handles human proximity override flags

► robot/heartbeat : Streams regular system alive pulses

► phoenix/cmd/move : Carries atomic teleoperation movement tokens

► phoenix/cmd/water : Controls high-pressure booster pump activations

► nav/target : Transmits Cartesian target positions

► phoenix/status : Compiles hardware operational feedback diagnostics|

By decoupling these communication pathways into individual topics, the interface handles simultaneous tasks—like streaming navigation goals, parsing fire detection confidences, and receiving raw system logs—without blocking the main user interface loop.

5.4. Real-Time Telemetry & Watchdog Monitor

To safeguard against communication losses while navigating high-hazard environments, the frontend incorporates an automated system-health engine.

1. Software Heartbeat Watchdog Implementation

The platform continuously streams status signals to the robot/heartbeat channel. The browser app captures these timestamps and feeds them into an active verification loop. If network degradation occurs or a hardware failure blocks the robot's transmission for more than a set threshold, the watchdog updates the user interface, flashes an alert flag, and triggers an audio alarm block to inform the safety officer.

```
// Hardware connectivity watchdog calculation loop
function evaluateNetworkLiveliness() {
  if (!lastHBTime) return;
  const latencySeconds = Math.floor((Date.now() - lastHBTime) / 1000);


  if (latencySeconds > 5) {
    $('hbTxt').textContent = `TIMEOUT: ${latencySeconds}s AGO`;
    $('pillHB').className = 'status-pill danger';
    $('hbDot').classList.add('critical-alarm-flash');
  } else {
    $('hbTxt').textContent = `ACTIVE: ${latencySeconds}s DELAY`;
    $('pillHB').className = 'status-pill ok';
    $('hbDot').classList.remove('critical-alarm-flash');
  }
}
```

2. Vector-Based Sparkline Visualization Engines

Rather than using heavy charting frameworks that increase rendering lag, the interface utilizes light HTML5 canvas components to draw trend data. Telemetry histories are stored in a continuous ring buffer capped at a strict maximum allocation threshold (MAX_PTS = 40).

$$\text{Horizontal Step Size} = \frac{\text{Canvas Width}}{MAX\_PTS - 1}$$

Every time a new data point arrives via robot/fire_alert or phoenix/status, the script calculates dynamic vector coordinates and maps them onto the canvas coordinate grid using inline Bézier control points:

```
// Custom vector path drawing loop for real-time telemetry sparklines
function renderLiveSparkline(canvasId, historicalDataPoints) {
  const canvas = $(canvasId);
  const ctx = canvas.getContext('2d');
  const width = canvas.offsetWidth;
  const height = canvas.offsetHeight;

  ctx.clearRect(0, 0, width, height);
  if (historicalDataPoints.length < 2) return;

  ctx.beginPath();
  const step = width / (MAX_PTS - 1);

  // Vector trajectory plotting loop
  historicalDataPoints.forEach((point, index) => {
    const xCoord = index * step;
    const yCoord = height - ((point / 100) * height);
    if (index === 0) ctx.moveTo(xCoord, yCoord);
    else ctx.lineTo(xCoord, yCoord);
  });
  ctx.strokeStyle = 'var(--fire)';
  ctx.lineWidth = 2;
  ctx.stroke();
}
```

5.5. Manual Override Engine & Keyboard Teleoperation Primitives

A critical safety feature of the dashboard is the **Manual Override System**, which allows operators to instantly supersede autonomous navigation loops. When the manual override switch is toggled, the application sends an authorization payload to the broker and unlocks a direct steering interface.

1. Keycode Event Mapping and Velocity Tokens

To ensure rapid, instinctual teleoperation without requiring manual screen interactions, the system binds core movement commands to keyboard event listeners. The handler maps characters directly onto standardized movement primitives:

```
// Keycode-to-hardware primitive control mapper matrix

const controlPrimitiveMap = {

  w: 'FORWARD',  ArrowUp:    'FORWARD',

  s: 'BACKWARD', ArrowDown:  'BACKWARD',

  a: 'LEFT',     ArrowLeft:  'LEFT',

  d: 'RIGHT',    ArrowRight: 'RIGHT',

  ' ': 'STOP'

};
```

2. Pulse Interrupt Safety Strategies

To prevent continuous command transmissions from clogging the local Wi-Fi router, the system utilizes a edge-triggered pulsing strategy. When an operator presses and holds a directional key, a single atomic payload (e.g., FORWARD) is transmitted to the phoenix/cmd/move channel. The hardware continues executing this vector until a keyup event intercepts the button release, instantly dispatching a STOP command to lock the motor gearboxes:

```
// Non-blocking interrupt keyboard control listeners

document.addEventListener('keydown', event => {

  if (!isManualModeActive || event.target.tagName === 'INPUT') return;

  const drivingToken = controlPrimitiveMap[event.key];

  if (drivingToken) {

    event.preventDefault();

    publishMqttMessage('phoenix/cmd/move', drivingToken);

  }

  if (event.key === 'Enter') publishMqttMessage('phoenix/cmd/water', 'ACTIVATE_PUMP');

});



document.addEventListener('keyup', event => {

  if (!isManualModeActive || event.target.tagName === 'INPUT') return;

  if (['w', 'a', 's', 'd', 'ArrowUp', 'ArrowDown', 'ArrowLeft', 'ArrowRight'].includes(event.key)) {

    publishMqttMessage('phoenix/cmd/move', 'STOP');

  }

});
```

This structural architecture ensures that if the browser tab loses focus or connection drops mid-maneuver, the motor control loop automatically drops to a neutral state, preventing the vehicle from moving out of control.

3. Coordinate Target Dispatch Interface

When semi-autonomous navigation is preferred over direct manual steering, the dashboard allows operators to send precise workspace targets. The operator inputs raw Cartesian location points (x and y coordinate boxes) and hits the dispatch button. The web application packages these inputs into a clean string layout and sends it to the central navigation stack over the network:

```
// Encapsulating Cartesian path goals into JSON formats

function dispatchNavigationTarget() {

  const targetX = parseFloat($('navX').value) || 0;

  const targetY = parseFloat($('navY').value) || 0;

  const targetPayload = JSON.stringify({ x: targetX, y: targetY });



  publishMqttMessage('nav/target', targetPayload);

  addLogEntry(`Nav2 Goal Dispatched: Coordinate Spatial Grid (${targetX}, ${targetY})`, 'info');

}
```

5.6 Cybersecurity Constraints, Vulnerability Profiles, and Production Hardening Requirements

Because Project Phoenix operates as a safety-critical fire suppression asset, the integrity, availability, and confidentiality of its bi-directional command telemetry stream are of paramount operational importance. In its current academic prototyping configuration, several explicit security limitations exist to simplify cross-platform testing and cloud hosting validation on the Vercel architecture.

The primary vulnerability profiles within the current proof-of-concept pipeline include:

1. **Unencrypted Transport Channels:** The system utilizes a standard, unencrypted Mosquitto MQTT broker transmission line. Passing atomic control payloads (e.g., FORWARD, ACTIVATE_PUMP) in plaintext leaves the network vulnerable to packet sniffing and malicious Man-in-the-Middle (MitM) command injection attacks.
2. **Absence of Perimeter Authentication:** The Vercel-hosted frontend deployment currently functions as a publicly open web endpoint. The lack of active user login constraints means any client possessing the URL could theoretically access teleoperation primitives or exploit MQTT credentials.

To transition this mobile platform from an academic prototype to a secure, resilient industrial production deployment, a three-tiered security hardening matrix must be implemented:

5.6.1 Transport Layer Hardening via MQTTS and X.509 Certificates

The baseline standard unencrypted TCP broker link must be completely deprecated in favor of secure MQTT (MQTTS) over TLS (Transport Layer Security). This protocol encapsulates all bi-directional JSON payload streams inside an encrypted cryptographic tunnel.

To prevent rogue hardware nodes from subscribing to sensitive fire alert topics or publishing unauthorized drive commands, strict mutual authentication (mTLS) must be enforced. This requires provisioning the onboard Raspberry Pi core and the hosting server with unique, cryptographically signed X.509 digital certificates validated by a private Certificate Authority (CA).

5.6.2 Role-Based Access Control (RBAC) and Token Authentication

The public presentation layer hosted on Vercel must be secured behind an enterprise identity provider utilizing OAuth 2.0 or secure JSON Web Tokens (JWT). Upon successful multi-factor authentication (MFA), an operator receives a time-restricted, cryptographically signed token.

The web application passes this token within the connection headers to authorize control. Furthermore, a strict Role-Based Access Control (RBAC) matrix must partition user permissions:

- **Auditor Role:** Granted read-only permissions to observe the HTML5 canvas telemetry sparklines and stream diagnostics.
- **Safety Incident Commander Role:** Granted full write permissions to transmit coordinate target dispatches, activate keyboard teleoperation listener primitives, and trigger emergency pump overrides.

5.6.3 Network Isolation via Private VPN Tunneling

To eliminate exposure to the public internet, the web interface, local message broker, and physical edge robot must be isolated within an encrypted Virtual Private Network (VPN) or a software-defined perimeter overlay (such as WireGuard or an industrial OpenVPN appliance). Under this architecture, the Vercel frontend acts as an internal network node communicating over an isolated subnet.

By closing public port access and routing all data through an encrypted VPN tunnel, the system removes external attack vectors, effectively neutralizing unauthorized scanning, spoofing, and Distributed Denial of Service (DDoS) attempts against the robot's edge core.

6. Chapter 6: Experimental Results & Testing Evaluation

6.1 Locomotion & Maneuverability Field Trials

To empirically validate the 4-Wheel Drive (4WD) skid-steer locomotion dynamics modeled in Chapter 3, a series of physical maneuverability trials were conducted on a smooth, industrial-grade indoor epoxy warehouse floor boundary. The testing layout focused on quantifying positional drift, tractive slip, and the accuracy of the inverse kinematics engine under varying fluid payload capacities.

6.1.1 Measured Turning Radius and Zero-Radius Pivot Verification

The kinematics model assumes that applying equal and opposite voltage cycles to the left and right motor banks yields an absolute zero-radius pivot turn ($R_t = 0.0$ mm) around the geometric center of the chassis. To test this, the robot was commanded via the Vercel manual override interface to perform ten continuous 360° rotations at a fixed PWM duty cycle of 65%.

The real-world deviation of the center of mass (CoM) was tracked using a floor-mounted grid and an overhead reference camera checking an attached ArUco fiducial marker.

The empirical field data compiled across payload states are detailed in the matrix below:

Test ID	**Fluid Payload Weight**	**Commanded Maneuver**	**Measured Turning Radius (Rt)**	**Maximum Centerline Drift**
L-01	0.0 kg (Empty Container)	360° In-Place Pivot	4.2 mm	±6.5 mm
L-02	2.0 kg (50% Fluid Load)	360° In-Place Pivot	7.5 mm	±9.2 mm
L-03	4.0 kg (100% Full Load)	360° In-Place Pivot	11.8 mm	±14.1 mm

The slight increase in the turning radius under full fluid load (11.8 mm) is directly linked to the lateral friction forces generated by the 130 mm rugged tires sliding sideways across the epoxy coating. However, an absolute displacement deviation of less than 12 mm confirms that the zero-radius pivot capability is operationally valid for navigating narrow warehouse pathways.

6.1.2 Empirical Slip Coefficient Validation

In Section 3.3.2, an empirical slip correction factor (C = 0.85) was introduced to reconcile the theoretical kinematic velocities with the lateral friction realities of the warehouse floor. To calibrate this parameter, the robot was driven along a linear 5.00 m reference track at a steady velocity input of 0.80 m/s.

The actual linear velocity (v_{actual}) was measured using a time-of-flight optical gate array, and the effective slip factor was calculated via the linear kinematics ratio:

$$c_{\text{measured}} = \frac{v_{\text{actual}}}{v_{\text{theoretical}}}$$

- **Theoretical Expected Life Translation:** At a 100% full load payload profile, the uncorrected theoretical model predicted a completion timeline of exactly 6.25 seconds (5.0 m / 0.8 m/s).
- **Real-World Empirical Results:** The prototype completed the track in 7.28 seconds under a 100% full fluid load, which matches an actual linear velocity of 0.687 m/s.
- **Resolved Correction Value:** Computing the ratio yields $c_{\text{measured}} = 0.687/0.80 \approx 0.858$.

This experimental result shows an exceptionally close correlation (<1% error distribution) with our initial theoretical calibration estimate (c = 0.85). This provides strong mathematical justification for the validity of the non-ideal kinematic equations included in Chapter 3.

6.2 Fluid Dynamics & Suppression System Field Verification

Active suppression validation testing focused on comparing the theoretical exit velocities derived via Bernoulli's theorem against the real performance of the Aqua Power AP 100 GPD booster pump and the 2-DOF targeting gimbal mechanism.

6.2.1 Measured Fluid Stream Jet Projection Range

The hydrodynamic framework in Section 4.3 calculated a theoretical fluid exit velocity of 35.23 m/s based on a nozzle working pressure of 90 PSI, translating to an ideal horizontal throw distance under zero-drag gravity vectors:

$$R_{\text{theoretical}} = \frac{v^2 \cdot \sin(2\alpha)}{g}$$

Setting a standardized elevation angle ($\alpha = 15°$) via the MG996R tilt servo, the ideal calculation predicts a maximum horizontal projection range of

$$R_{\text{theoretical}} = (35.23)^2 \cdot \sin(30°)\ /9.81 \approx 63.26\ \text{m}$$

Physical testing was performed in an isolated testing area, with the pump operated across a spectrum of manual pressure levels monitored via an inline analog pressure gauge. The actual water jet range was measured from the front edge of the nozzle to the primary pool impact center line:

Operating Pressure	**Ideal Range (Rtheoretical)**	**Measured Stream Range (Rexperimental)**	**Calculated Stream Efficiency**
30 PSI	21.08 m	4.15 m	19.68%
60 PSI	42.17 m	6.82 m	16.17%
90 PSI (Nominal Working)	63.26 m	8.45 m	13.35%

6.2.2 Hydrodynamic Error Analysis

The massive variance between the theoretical model (63.26 m) and the actual measured stream range (8.45 m) highlights the limitations of using a basic, frictionless Bernoulli calculation for ambient air projections. As established in Section 4.3.1, while internal hose friction losses are quite low ($\Delta P_{\text{loss}} \approx 0.031$ PSI), the external stream undergoes immediate atomization upon leaving the nozzle tip.

The continuous high-pressure jet stream breaks apart into discrete droplets, dramatically increasing its effective surface area and exposing the liquid payload to intense aerodynamic drag forces ($F_d \propto v^2$). This mechanical energy loss clamps the real deployment range to 8.45 m. However, an 8.45 m operational envelope is fully sufficient for indoor warehouse aisle deployment, allowing the robot to suppress flame zones from a safe distance without exposing its onboard electronics to high radiant heat flux.

6.3 Network Latency & Communication Infrastructure Audits

To confirm that the web-based remote control system remains reliable during safety-critical overrides, the communication loop between the Vercel cloud environment and the local Mosquitto MQTT broker network was thoroughly benchmarked.

The execution loop latency was evaluated by embedding a precise millisecond timestamp within a diagnostic telemetry packet published to the phoenix/ping channel. The onboard Raspberry Pi edge script was configured to instantly echo the exact payload back through a dedicated channel (phoenix/pong). The dashboard intercepted the return signal, calculated the total delta time, and rendered the moving tracking average onto the console log framework.

Tests were performed across 500 consecutive command loops to compile a realistic performance profile:

- **Minimum Registered Latency:** 14 ms (Under ideal local network routing states).
- **Maximum Registered Latency Spikes:** 92 ms (Caused by periodic Wi-Fi signal attenuation and buffer shifts).
- **Mean System Latency (Tracking Average):** 38 ms.

A mean round-trip latency of 38 ms matches the design criteria for responsive real-time teleoperation. Because human cognitive reaction time is roughly 150 ms – 200 ms, the control pipeline updates fast enough to feel instantaneous to a human operator navigating via keyboard listeners.

6.4 Full System Integration: Mock Deployment Scenario

To demonstrate the full integration of the mechanical frame, active fluid electronics, and cloud communication infrastructure, a mock deployment trial was conducted within an indoor warehouse test corridor.

The integrated execution path followed a structured operational sequence:

1. **System Initialization:** The robot was placed at the entry boundary of an aisle corridor. The onboard Raspberry Pi established network connection with the local Mosquitto broker, and the dashboard panel transitioned to MQTT LIVE.

2. **Autonomous Traversal:** The autonomous navigation script was launched, commanding the 4WD skid-steer chassis to move down the corridor at an operating speed of 0.68 m/s while updating live location tracking data streams.
3. **Manual Override Interception:** A controlled ignition source was activated at a distance of 6.00 m from the vehicle pathway. The operator toggled the manual override switch on the Vercel frontend, instantly blocking autonomous control loops and gaining direct teleoperation access within 38 ms.
4. **Targeting Alignment:** Using the edge-triggered keyboard listeners (WASD), the safety officer executed an in-place zero-radius turn to align the front chassis profile with the fire hazard area.
5. **Active Fire Suppression:** The operator pressed the Enter key on the control station interface, sending an immediate activation signal to the phoenix/cmd/water channel. The onboard relay triggered the 24V Aqua Power booster pump, projecting an accurate high-velocity fluid stream across a distance of 8.45 m to successfully douse the flame source.

The full mission sequence was documented via an external high-definition video camera, providing clear empirical proof that the integrated mechanical, fluid, and digital systems can perform reliable emergency teleoperation.

7. Conclusion

This thesis has presented the comprehensive mechanical design, structural fabrication, fluid dynamics engineering, and full-stack teleoperation infrastructure of the Project Phoenix mobile firefighting platform. By shifting the core research focus away from high-latency external computing architectures, this investigation successfully established a robust physical execution envelope paired with a highly responsive, low-latency human-in-the-loop control framework. The structural skeleton, built upon an interconnected space-frame of 2020 V-slot aluminum extrusions and reinforced with custom-cut CNC sheet metal panels, successfully demonstrated high torsional rigidity and fluid-splash protection. Additive manufacturing utilizing impact-modified Creality Hyper PLA on an Elegoo Neptune 3 Pro system enabled the rapid development of complex geometric components, including the 2-DOF suppression nozzle gimbal and a 4-liter internal liquid reservoir sealed hermetically against hydrostatic weeping via a low-viscosity clear epoxy resin lining. Locomotion parameters were thoroughly validated through a 4-Wheel Drive (4WD) skid-steer configuration powered by four planetary-geared JGB37-545 DC motors, which successfully executed rapid, in-place zero-radius turns on indoor epoxy floor boundaries after resolving output shaft tolerance mismatches via precision-drilled 6mm interference couplers. Active fire suppression was fully realized through the integration of an updated 24V Aqua Power AP 100 GPD booster pump capable of generating high-velocity jet streams up to 150 PSI. Finally, the cloud-hosted web control dashboard deployed on Vercel utilized a decoupled, asynchronous Mosquitto MQTT message broker to empower remote operators with real-time vector-based telemetry sparklines and an edge-triggered keyboard manual override engine with sub-second feedback loop latency.

Despite the successful implementation of the prototype system, several engineering boundary conditions and limitations were identified during active validation testing. Mechanically, the direct motor-to-chassis rigid mounting configuration lacks active damping, meaning the platform is highly optimized for smooth, indoor warehouse epoxy floor coatings but is vulnerable to wheel slip or tread vibration if introduced to rugged, unpaved terrains. Hydraulically, while the ideal Bernoulli model accurately reflects stream profiling at low internal flow velocities, the discharge stream faces severe aerodynamic drag friction once it leaves the nozzle exit plane, causing predictable velocity degradation over extended distances. Furthermore, utilizing a commercial-grade reverse osmosis booster pump introduces duty cycle limitations; although highly cost-effective and fully capable of sustaining the required 150 PSI across the prototype's 2.22-minute total fluid payload depletion window, it cannot operate continuously for prolonged periods without risking thermal saturation.

To expand upon the physical and digital frameworks established across this design iteration, several critical future work pathways are recommended. First, integrating independent passive or active suspension joints at each motor corner module would dramatically stabilize traction dynamics when crossing uneven threshold transitions or debris fields. Second, replacing the aluminum extrusion framework with lightweight high-tensile carbon composite tubes would optimize the platform's strength-to-weight ratio, effectively expanding the available payload capacity for fluid storage without drawing additional motor torque or current. Finally, at the software architecture layer, migrating the web telemetry data transfer pipelines from basic MQTT clients over to an HTTP/2 Server-Sent Events (SSE) framework would minimize communication protocol overhead, guaranteeing highly efficient, real-time diagnostic rendering on the canvas engine even over severely congested industrial wireless connections.

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